

Deferred Settlement, v16-Native

The open window between trade-date recognition and custody movement —
deferred-settlement design cell, converged Round 10, seven approvals 2026-07-16

1 The Problem

Equity markets settle on a delay. On Monday at 14:32:11 UTC a portfolio wallet w buys 100 XYZ at \$50.00 — \$5,000.00 against 100 shares, T+2, settling Wednesday. The trade is economically complete on Monday: exposure held, cash owed, dividends due from that instant. The custody fact happens on Wednesday, when the depository moves the shares and the cash agent debits the nostro. Between the two lies the open settlement window $[T, t_d]$; its length is a parameter — T+1, T+2, T+0, T+5 on a cross-border fail — never a doctrine.

The window makes a number real before anything moves. XYZ closes at \$52.00 on Tuesday. The book must show +\$200.00 of unrealised profit at Tuesday's close — recognised, capital-bearing, reported — while the custodian still reports \$1,000,000.00 and zero XYZ, exactly as on Monday morning. Both records are right: different facts, different calendars, different authorities. Collapse them and the system fails one way or the other: debit the nostro at T and the statement breaks by \$5,000 all window with nothing left to absorb Wednesday's real debit; recognise nothing until t_d and Tuesday's \$200 does not exist — profit and loss, risk, and capital misstated. The design must keep both facts, separately observable and separately attested, bound by an identity computable inside the system. A bound living in prose is checked in a spreadsheet, and the spreadsheet is a second store of truth — the defect this ledger abolishes.

v16.0 already carries the machinery: trade-date booking as a declared term, the settlement-obligation unit with its `instructed/settled/failed` node, the partial edge, market claims, the event-kind registry, the bitemporal doctrine. What it lacked was the bound (gap NS-01). This document is that bound.

2 The Design

Everything below is recorded events and folds of them. No state exists outside the log: no mirror wallet, no stored settlement status, no stored break register, and the execution substrate owns no fact.

2.1 Events: the witness kinds

External assertions cross the one door as registered, typed, moveless observations — provenance carried, quarantined when malformed, superseded only at the tip of a per-(account, as-of) chain, never edited. Three asserter classes, one trust assumption (TA-CUSTODY, §5):

- **Balance statements** (camt.053, the depot statement): per account and unit, a balance at an as-of time; fields account, coordinate, balance in integer minor units, as-of, statement-id, supersedes-ref. An unresolvable account quarantines and re-admits when the reference resolves. Each account declares a cadence on its venue calendar; a missed cadence mints an overdue item — a dark custodian never leaves a stale CLEAN standing. Statements are authenticated under an identified key; keys live in a valid-timed registry, and a revocation carries a compromise-effective time t_c scoping re-quarantine deterministically.
- **Settlement and cash confirmations** (sese.025, camt.054), including partial clips, always recorded — a clip below the venue's declared minimum fill size accumulates on the one live residual instead of splitting it.
- **Penalty advices and emission receipts** (semt.044; dispatch-intent, dispatch receipt or CSD acknowledgment, match confirmation with its matching timestamp, cancellation confirmation). The dispatch-intent is recorded *before* the send — the egress dual of witness-before-signal — and keys the repair discriminator. The dispatch receipt is an internal-egress witness (our gateway attesting our own send), its detection indirect. The matched date and timestamp are CSD-asserted, single-source:

bounded and confronted, never verified.

2.2 The fold

Each event kind folds into a fixed consequence. Instruction: the paired **owned** re-book and the moveless minting of the settlement-obligation unit at **instructed** (legs, due date, positive inherited minimum fill size, root settlement-instruction reference, per-unit settlement channel) — on a declared atomic-finality venue (a TA-TERMS term) the unit is born **settled** and no timer is armed, so the T+0 race is unrepresentable. Confirmation: the moveless walk to **settled**. Partial clip at or above the floor: the certified split — settled leg plus one live residual, root reference inherited. Fail notice, or the due-date watch reading cumulative confirmed quantity short of instructed: the walk to **failed** and the fails cascade. Repair event: in-place repair when no dispatch-intent is recorded; otherwise cancel-and-re-instruct on the recorded cancellation confirmation, one atomic transaction. Recall: a paired lent-coordinate move plus a possession-plane return leg; a terminally failed return routes to close-out, the defaulter’s wallet receiving the securities it cannot return, paired with the lender’s write-off, a valued claim minted against collateral. Custody statement: no balance change — the reconciliation projection reads it. Reconciled penalties book, per (participant, currency, period), an **owned** charge plus a penalty settlement-obligation unit discharged by the collection advice. A buy-in delivery discharges the failed residual it compensates; it never books a second long.

CSDR penalties fold over dated confirmation clips keyed by root reference, on sequential windows: late-matching over $[\text{ISD}, M)$ or $[\text{ISD}, M]$ as the matching timestamp falls before or after the venue’s declared settlement cut-off on day M ; settlement-fail over $[\max(\text{ISD}, M), \min(\text{settlement}, \text{buy-in}, \text{cancellation}))$ — ISD inclusive, all right endpoints exclusive, no day charged under both. Each advice line passes three decidable checks: its own arithmetic, exact (quantity $\times$ its reference price $\times$ rate = its amount); its reference price against that business day’s recorded close from the SECR-designated source, within a governed de-minimis tolerance; its failing quantities against our clip fold. CLEAN or BREAK — no fuzzy middle class. A lateness charge against us is confronted with the matched instruction’s own send-completion witness; if that witness is absent from the log, the advice line quarantines as an open item, never a silent CLEAN.

2.3 The invariants

Invariant 1 (The reconciliation identity). *For every custody account a , unit u , and recorded statement with as-of time t , the residual*

$$r(a, u, t) = \text{stmt}(a, u, t) - \sum_{w \in W(a)} [\text{owned} - \text{lent_out} + \text{borrowed} + \text{inflight_out} - \text{inflight_in}](w, u, t)$$

is zero within the tolerance declared for (a, u) . W maps each wallet’s mass to exactly one account per coordinate class (total, disjoint, covering); the in-flight sums run over the directed legs of w ’s unsettled settlement-obligation units. The three-way reading is a report projection, not part of the invariant: CLEAN iff $r \approx 0$ and the in-flight explanation $F = 0$; EXPECTED-LEAD-LAG iff $r \approx 0$ and $F \neq 0$, each line carrying the obligation-liveness status of its naming units; BREAK iff $r \neq 0$, identified by (account, coordinate, residual sign), recomputed from the log, never stored, repaired forward only.

Invariant 2 (The Spine: the as-of read-set). *Everything Invariant 1 reads — balances, in-flight legs, the account map W , the tolerance schedule, the signing key — folds in force at the statement’s as-of time. Registry facts, tolerances, and keys are recorded declared data with valid time, amended forward only. A past classification is never rewritten by a later amendment; a revoked key re-quarantines exactly the statements admitted in $[t_c, \text{discovery}]$ under it.*

Invariant 3 (The Binding: one-parameter construction). *Every transaction that creates or changes a settlement-obligation unit’s legs derives the legs and the balance moves from one restated parameter set,*

so a unit whose legs disagree with its booking cannot be constructed and $\Delta r = 0$ is a theorem of the transaction shape. Successors net recorded settled mass per coordinate, at the amounts actually moved. The forward-or-return decision is taken independently per coordinate: forward leg of restated – settled where restated $\geq$ settled; return leg of settled – restated, opposite direction, where it is not. All legs are positive. Legs partition into units by external instruction — each unit’s legs belong to exactly one root reference (a function, not a bijection: one reference carries several units only through the certified partial split, whose inheritance the penalty fold depends on). Whether a mixed correction settles netted or gross is a declared term of the agreement, and both conserve identically. In numbers (price corrected to \$10.00 over 30 settled at \$50.00): netted, receive 70 free of payment plus a \$500.00 cash return; gross, delivery-versus-payment 70 against \$700.00 plus a \$1,200.00 compensation — either way predicted *nostro* is $999,000 + \text{out} - \text{in} = 998,500.00$ mid-window and total cash paid ends at the restated \$1,000.00.

Invariant 4 (Terminality: settled mass is terminal). *Recorded settled mass is never retired, never negated, never raced. Successors net it; corrections below it unwind it forward through return obligations; a settlement confirmation racing a pending cancellation resolves in settlement’s favour, and the pending repair re-routes to a post-settlement compensating transaction — never a hung amendment.*

2.4 Temporal mapping

The workflow is the settlement-obligation unit’s own unit-workflow, started by signal-with-start on the recorded instruction; the record is primary, and a wiped substrate re-derives every workflow, timer, and in-flight quantity from the log. Activities are the four side-effecting steps: capture a witness through the door (before any signal exists), evaluate the unit’s contract, propose to the door, emit the instruction. The timer firing at t_d is the due-date watch’s liveness half and carries no data: it wakes the unit; the recorded quantities and the contract decide. A witness signal carries only the log position of an already-recorded fact — a lost signal costs latency, never the record. Egress runs intent, send, receipt capture; emission idempotence rides (unit, node), never a substrate identifier. A split completes the parent workflow and starts one per successor; an atomic-finality unit arms no timer at all.

3 Worked Example

Buy 100 XYZ at \$50.00; T Monday, t_d Wednesday ($T+2$); closes \$50.00 / \$52.00 / \$51.50. Pre-trade: owned(USD) = 1,000,000.00, owned(XYZ) = 0; statements: *nostro* 1,000,000.00, depot 0; both accounts CLEAN ($r = 0$, $F = 0$); $V = 1,000,000.00$.

Monday, T . One transaction: move owned(XYZ) 100, seller $\rightarrow w$; move owned(USD) 5,000.00, $w \rightarrow$ seller; the unit SO-1 minted at **instructed** (deliver 100 in, pay 5,000.00 out, due Wednesday). Conservation: $\Sigma_{XYZ} = +100 - 100 = 0$; $\Sigma_{USD} = -5,000 + 5,000 = 0$. Balances: owned XYZ 100, USD 995,000.00. $V_T = 995,000 + 100 \times 50.00 = 1,000,000.00$: a fair exchange, zero profit and loss at the trade instant. Identity — *nostro*: $995,000 + 5,000 - 0 = 1,000,000 =$ statement, $r = 0$, $F \neq 0$: EXPECTED-LEAD-LAG, named by SO-1. Depot: $100 + 0 - 100 = 0 =$ statement: EXPECTED-LEAD-LAG.

Tuesday close, \$52.00. No moves; conservation vacuous. $V_{T+1} = 995,000 + 100 \times 52.00 = 1,000,200.00$: profit and loss +\$200.00, unrealised, with zero cash movement and zero custody movement. The custodian still reports 1,000,000.00 and 0; the identity still balances through SO-1’s in-flight legs. The \$200 is real, the custody silence is real, and the record carries both without contradiction.

Wednesday, finality. sese.025 and camt.054 cross the door as moveless observations; SO-1 walks **instructed** $\rightarrow$ **settled** — zero moves, conservation vacuous, owned untouched. Statements: depot 100, *nostro* 995,000.00. Identity: $995,000 + 0 - 0 = 995,000$ and $100 + 0 - 0 = 100$: CLEAN, CLEAN. $V_{T+2} = 995,000 + 100 \times 51.50 = 1,000,150.00$; settlement contributed zero to V at every step.

Fail and recover (buy-in). Monday and Tuesday as above; the +\$200.00 stands. Wednesday: no confirmation; the due-date watch fires; cumulative confirmed $0 < 100$; SO-1 walks to **failed** on the full 100. Owned untouched, statements unchanged, legs still in-flight under the unsettled unit:

both accounts read $r = 0$, $F \neq 0$ — EXPECTED-LEAD-LAG, its naming unit printed as “SO-1 failed, overdue 1 bd”, never a false BREAK. Thursday a buy-in executes at the \$51.50 mark. Its delivery confirmation discharges SO-1 (settled-by-buy-in): no second long; owned XYZ stays 100. One compensating cash transaction, three paired moves: 5,000.00 seller $\rightarrow w$ (the unearned trade-date payment returns); 150.00 seller $\rightarrow w$ (the differential, borne by the failing party); 5,150.00 $w \rightarrow$ buy-in counterparty. $\Sigma_{\text{USD}} = +5,000 + 150 - 5,150 = 0$ on w : owned(USD) remains 995,000.00 — the buyer paid exactly \$5,000.00 for its 100 shares; the \$150.00 landed on the failing seller. Once the buy-in settles: depot 100, nostro 995,000.00, CLEAN, CLEAN; the buy-in date closes the penalty window.

4 What Was Considered and Rejected

- *Debit the nostro at T* . The statement breaks by \$5,000 all window; Wednesday’s real debit has nothing to discharge.
- *Recognise nothing until t_d* . Tuesday’s \$200 does not exist; trade-date accounting, risk, and capital misstated.
- *Mirror wallets* (the v11 contras and depot/nostro mirrors): our own copy of the custodian’s book — a second store that itself needs reconciling. The unit’s legs and the recorded statement replace both.
- *Stored obligation rows, break registers, materialised statuses*. All computable from the record; storing them re-admits divergence.
- *Multi-source quorum on statements*. A statement is not a price; a quorum over one custodian’s own account is theatre.
- *An in-flight coordinate on PositionState*. A projection stored twice.
- *A recursion depth cap on partial splits* (adopted Round 1, deleted Round 2: exactly one residual is ever live, and a cap drops witnessed confirmations). Replaced by the venue minimum fill size plus aggregation.
- *An “expected difference” band on penalty reconciliation* (adopted Round 4, retired Round 5): an ungoverned band masks a real overcharge. Replaced by three decidable checks.
- *An identity-excluded payable coordinate*: an identity with an exclusion list is an identity with a memorised hole.
- *A sixth trust assumption per message family*: a new named assumption is earned by a different proposition, never a different publisher.

5 Declaration

Rounds used: 10. Seven named reviewers (correctness-architect, minsky, jane-street-cto, nazarov, sbl-specialist, regulatory-reporter, rosetta-cdm); every break answered with a mechanism or a counterexample; four interim constructions falsified by review and corrected in print (the depth cap, the parallel penalty windows, the single-sided write-off, the bundled mixed-case successor). Round 10: seven approvals, each with an attack record.

DS1–DS19 disposition (the v11 acceptance checklist; none silent):

DS	Invariant	Disposition
DS1	Economic exposure at T	Satisfied: owned re-books at instruction; no settlement event moves it.
DS2	Conservation	Satisfied: per (unit, coordinate) at the one door; no carve-out.
DS3	Reconciliation identity	Satisfied: Invariant 1, closed by this design (gap NS-01).
DS4	No discharge without witness	Satisfied: node walks only on recorded triggers; no clip is ever dropped.
DS5	Replay determinism	Satisfied: Spine + tip-only supersession; same log, same classification.
DS6	Idempotent finality messages	Satisfied: cause-derived identity at the door.
DS7	Failure non-reversal	Satisfied: failure is a recorded event; unwind is compensating, forward.
DS8	Status monotonicity	Satisfied: property of the unit’s certified product graph.

DS9	Buy-in compensation closure	Satisfied: trade plane by the binding; possession plane by close-out routing.
DS10	Cross-currency Herstatt	Subsumed: routed to the NS-02 currency workstream (interface below).
DS11a	Partial per-step conservation	Satisfied: certified split; $q_s + q_r = Q$; parent retires from zero.
DS11b	Partial monotonicity, bound	Satisfied: positive inherited fill-size floor + aggregation (closes NS-04).
DS12	Variant degeneration T+0..T+5	Satisfied: the cycle is a declared term; atomic finality births the unit settled.
DS13	—	Not defined in the source checklist (v11 numbering gap); nothing to disposition.
DS14	Penalty schema determinism	Satisfied: both CSDR penalties, partitioned windows, decidable checks, booked cash leg.
DS15	Default close-out	Satisfied: certified close-out; the fails cascade routes into it.
DS16	Bitemporal restatement	Satisfied: longer log, never an edit; supersession tip-only.
DS17	Capability-scoped status writes	Satisfied: one writer, the unit's own contract.
DS18	DvP atomicity	Satisfied at ledger level, stated honestly: CSD atomicity reconciled, never guaranteed.
DS19	Witness-identity determinism	Satisfied: cause-derived identifiers; origin authenticity on statements.

Interfaces to queued problems. DS10's cross-currency legs and Herstatt projection go to the NS-02 currency workstream (an FX leg is two settlement-obligation units whose joint open state projects as exposure); the close-out claim's accrued rebate/fee/manufactured-payment income leg goes to the billing/income workstream (the claim is never the complete close-out amount); the executable-property set — identity, Binding constructibility, window partition, race interleavings, each with a required firing witness — goes to the test regime.

Decision log. DL-01 (panel, 3–0): the custodian statement crosses the one door as a registered witness event kind, and TA-CUSTODY is named — proposition: a recorded custody assertion faithfully renders its asserter's book; owner: data governance; violation: misrender, forgery, replay, silence; detection: Invariant 1 classifying BREAK, signature and sequence verification, the cadence watch; residual: the coincidental misrender, the send-to-CSD-entry gap, the matched-date assertion inside a genuinely late window. DL-02 (unanimous, 7–0): penalty advices and emission receipts ride under TA-CUSTODY as enumerated asserter classes — a new named assumption is earned by a different proposition, never a different publisher.

Deferred to the owner (spec-pass proposals, not constitutional amendments). Naming TA-CUSTODY as the fifth trust assumption edits one sentence of the v16.0 specification, Chapter 16: “The boundary rests on four named trust assumptions and no other” becomes “...five ...: TA-KIND, TA-TERMS, TA-EXDATE, TA-ARRIVAL, TA-CUSTODY,” with the four-part clause above appended. This is precisely a specification amendment: the constitution (C-4.12) names the trust-assumption pattern and fixes no count; only the specification sentence enumerates four. The venue cut-off's faithfulness rides the existing TA-TERMS. Both edits are parked with exact text for the owner's spec pass.

Confidence. HIGH on the design: ten adversarial rounds, seven reviewers, every mechanism individually attacked, all residual trust honestly named. The one open obligation is execution: the property set is specified with required firing witnesses but has not yet run, and a property that has never fired witnesses nothing. Confidence becomes final when the witnesses fire.